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BCA (Sem-IV) – Design & Analysis of Algorithm (401) Core-VIII

# 2025

*Time : 3 hours*

*Full Marks : 70*

*Candidates are required to give their answers in their own words as far as practicable.*

*The figures in the margin indicate full marks.*

*Answer all sections as directed.*

## Section–A

Objective type Questions

**Compulsory**

1. Choose the correct answer of the following multiple choice questions : 2×10=20

   (i) Iterative design technique mainly focuses on:

A. Designing algorithms using recursion only

B. Solving problems by repeating a set of steps

C. Dividing a problem into subproblems

D. Using backtracking approach

(ii) The three main steps of Divide and Conquer are:

A. Divide, Sort, Merge

B. Break, Solve, Combine

C. Divide, Conquer, Combine

D. Split, Iterate, Join

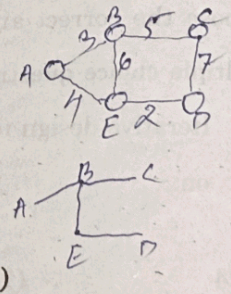
(iii) If a graph has V vertices, how many edges does its MST contain?

A. V

B. V+1

C. V–1

D. V–2






(iv) What is the best-case time complexity of insertion sort?

A. $O(n^2)$

B. $O(n \log n)$

C. $O(n)$

D. $O(\log n)$

(v) Merge sort follows which design technique?

A. Greedy method

B. Dynamic programming

C. Divide and conquer

D. Backtracking

(vi) Which sorting algorithm is generally the fastest in practice for large datasets?

A. Bubble sort

B. Insertion sort

C. Merge sort

D. Quick sort

(vii) BFS explores a graph by :

A. Going as deep as possible first

B. Exploring all neighbors before moving to next level

C. Selecting minimum weight edges

D. Using recursion only

(viii) DFS uses which of the following data structures?

A. Queue

B. Stack

C. Priority Queue

D. Heap

(ix) Which condition correctly represents moving to the right subarray in binary search?

A. arr[mid] > key

B. arr[mid] == key



C. arr[mid] < key

D. arr[mid] >= key

(x) In linear search, the maximum number of comparisons for an array of size n is

A. $n-1$

B. $n$

C. $n/2$

D. $\log n$

## Section–B

Short Answer type Questions

Answer any **four** questions of the following : 5×4=20

2. Write the Importance of designing an algorithm.

3. What is the Divide and Conquer algorithm design technique?

4. Write a short note on the working of Greedy algorithms.

5. Write an algorithm for insertion sort.

6. Write the working principle of BFS.

7. What are the features important for "Correctness of an algorithm"?

8. What do internal nodes and leaf nodes represent?

9. Why are decision trees used in DAA?

## Section–C

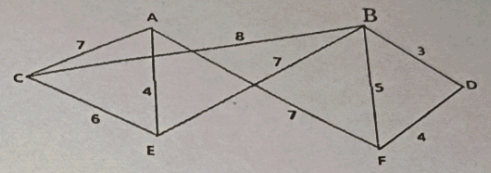
*Long Answer type Questions*

Answer any **two** questions of the following : 15×2=30

10. Calculate the time complexity of Bubble Sort in best and worst cases.



11. Find the Minimum Spanning Tree of the given graph using Prim's algorithm.




12. Analyze the time complexity of Binary Search in best and worst cases.

13. Write short note on: Dynamic programming, DFS and Sorting techniques.

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